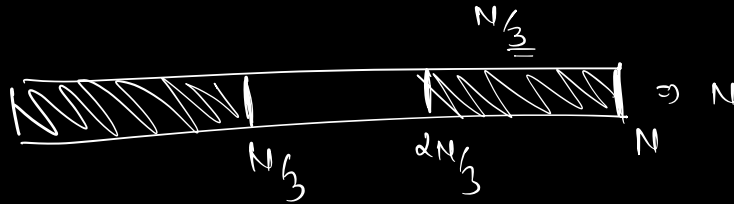


Q Majority element :- $\text{freq} > N/3$.



ex $\Rightarrow$

0	1	2	3	4	5	6	7	8	9	10	11	12
4	4	2	4	3	2	4	2	3	2	4	2	4
✓	✓		✓			✓				✓		✓

length $\Rightarrow 13$.

ME $\Rightarrow$ min freq $\Rightarrow > N/3 \Rightarrow \frac{13}{3} - 4 \Rightarrow \boxed{5}$

count of 4 $\Rightarrow 6$

length remain $\Rightarrow 13 - 6$
 $= \boxed{7}$

count 2 = 5.

at most — 2 ME can be present.

ex $\Rightarrow$ arr $\Rightarrow$

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
3	2	3	5	3	3	3	6	7	1	3	14	8	0	9
15	16	17	18	19										
1	3	7	6	12										

$N \Rightarrow 20$. $\longrightarrow$ ME ($> N/3$)

min freq $\Rightarrow 7$

Count of <u>3</u>	N	
7	20	$\rightarrow 7 > (20/3)$
6	17	$\Rightarrow 6 > (17/3)$
5	14	$\Rightarrow 5 > (14/3)$

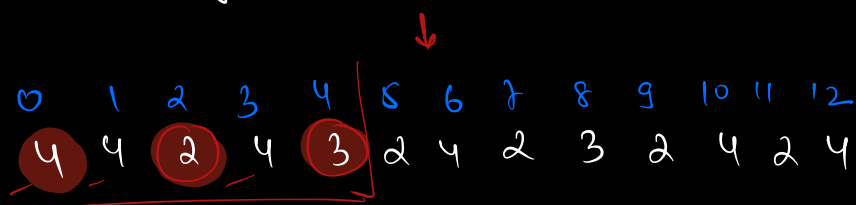
*** If we remove 3 distinct elements from the array, the ME remains the same

Since, max^m of 2 ME are possible

ME1	ME2
C1	C2

* find out best candidates for ME1 & ME2

* verify for both.



ME1 : 4

C1 : 2 3

ME2 : 2

C2 : 1

> N/4

C1 - -

C2 - -

SORTING

Agenda:-

- i) why we need sorting?
- ii) problems using sorting
- iii) comparator

: Sorting algo $\Rightarrow$ advanced

Sorting:- arranging data in any specific order based on a factor

of eCommerce

$\Rightarrow$ sort by price

$\Rightarrow$ sort by popularity

$\Rightarrow$ interview sort

: sort the problems by the toughness

: sort problems by companies

: deck of cards:-

i) By color

ii) By type

iii) By value

iv) By face value

⇒

3	8	9	14	19	20
---	---	---	----	----	----

 ⇒ sorted

↳ sorted in asc by value

⇒

19	14	9	8	3
----	----	---	---	---

 ⇒ sorted

↳ sorted in desc by value

⇒

1	13	9	6	12
---	----	---	---	----

 ⇒ sorted

factors

↓ ↓ ↓ ↓ ↓
1 2 3 4 6

↳ sorted in asc by no. of factors

⇒ language → sorting library functions

function ⇒ $O(N \log N)$ ⇒ TC

SC ⇒ depends on algo

$O(1) \rightarrow O(\underline{N})$

∴ Sorting helps us to search
things.

Q1. Given an array of size N , you can remove any one element at a time. Cost of removal $\Rightarrow$ sum of element in arr. before removal. Find the min cost to remove all elements.

$$A = 2 \quad 1 \quad 4$$

$$\text{remove} \Rightarrow 2 \Rightarrow 2 + 1 + 4 = 7$$

$$A = 1 \quad 4$$

$$\text{remove} \Rightarrow 1 \Rightarrow 1 + 4 = 5$$

$$A = 4$$

$$\text{remove} \Rightarrow 4 \Rightarrow 4 = 4$$

$$\begin{aligned} \text{total} \\ &= 7 + 5 + 4 \\ &= \underline{\underline{16}} \end{aligned}$$

ex $\Rightarrow A = 2 \quad 1 \quad 4$

$$\text{remove} \Rightarrow 4 \Rightarrow 2 + 1 + 4 = 7$$

$$A = 2 \quad 1$$

$$\text{remove} \Rightarrow 2 \Rightarrow 2 + 1 = 3$$

$$A = 1$$

$$\text{remove} \Rightarrow 1 \Rightarrow 1 = 1$$

$$\underline{\underline{11}}$$

Q. $\Rightarrow$

4, 6, 1

$$\text{remove } 6 \Rightarrow 6 + 4 + 1 = 11$$

4, 1

$$\text{remove } 4 \Rightarrow 1 + 4 = 5$$

$$\text{remove } 1 \Rightarrow 1 \Rightarrow 1$$

(17)

ans.

Q.

3 5 1 -3

$$\text{remove } 5 \Rightarrow 3 + 5 + 1 + (-3) = 6$$

$$[3 \ 1 \ -3] \text{ remove } 3 \Rightarrow 3 + 1 + (-3) = 1$$

$$[1 \ -3] \text{ remove } 1 \Rightarrow 1 + (-3) = -2$$

$$[-3] \text{ remove } -3 \Rightarrow -3 = -3$$

(2)

ans.

: generalise $[a, b, c, d]$

$$\text{remove } a \longrightarrow a + b + c + d$$

$[b, c, d]$

remove b

$$\longrightarrow$$

$$b + c + d$$

$[c, d]$

remove c

$$\longrightarrow$$

$$c + d$$

d

remove d

$$\longrightarrow$$

$$d$$

total cost $\Rightarrow a + 2b + 3c + 4d$

total cost $\Rightarrow$ a $+$ $2b$ $+$ $3c$ $+$ $4d$

$\downarrow$
minimize

$\downarrow$
 take a
 as the largest-
 value

$\downarrow$
 take d
 as the least-
 element

$a \xrightarrow{\hspace{10em}} d$
 Max Min

arr $\Rightarrow$

a_0
 $\downarrow$
 Max

a_1

a_2

a_3
 $\downarrow$
 Min

$\xrightarrow{\hspace{10em}}$ sorted in
 desc order

<u>a_0</u>	$\Rightarrow$	$a_0 + a_1 + a_2 + a_3$	PF [3]
a_1	$\Rightarrow$	$a_1 + a_2 + a_3$	PF [3] - PF [0]
a_2	$\Rightarrow$	$a_2 + a_3$	PF [3] - PF [1]
a_3	$\Rightarrow$	a_3	PF [3] - PF [2]

PFsum $\Rightarrow$

(i) sort()

(ii) find cost

(i) PFsum

$$TC \Rightarrow O(N \log N) + O(N) + O(N)$$

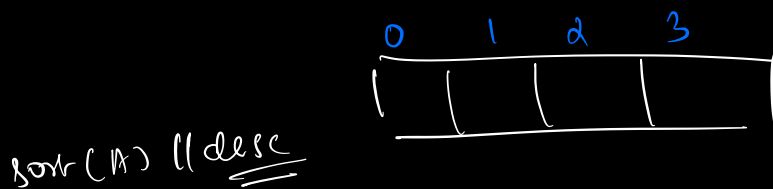
sort
Ptsum
cost

$$\Rightarrow O(N \log N)$$

$$SC \Rightarrow O(N)$$

Ptsum array

$\Rightarrow$ Contribution technique



$$\Rightarrow A[0] + 2A[1] + 3A[2] + 4A[3]$$

$$\hookrightarrow \underline{(i+1)A[i]}$$

Pseudo

cost = 0

sort(A); // in desc order

for (i = 0; i < N; i++) {

cost = cost + (i+1) * A[i]

}

return cost

}

$$TC \Rightarrow O(N \log N) + O(N)$$

$\downarrow$
 sort

$\downarrow$
 cost

$$\approx O(N \log N)$$

SC $\Rightarrow$ depends on sorting algo

* if sorting is a part of SDN

$\hookrightarrow$ library fun

* if asked to do some kind of sorting

$\hookrightarrow$ algo.

Amazon, Zeta

Q2. Noble integers

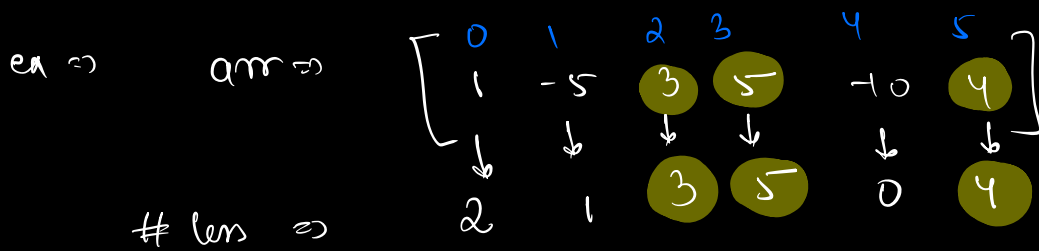
Given an array of size N. Count no. of noble integers present in array, array only contains distinct elements.

Noble int. :-

$A[i]$ is a noble int, if

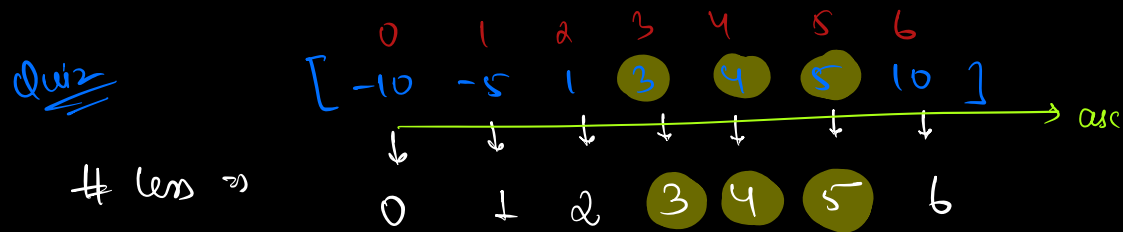
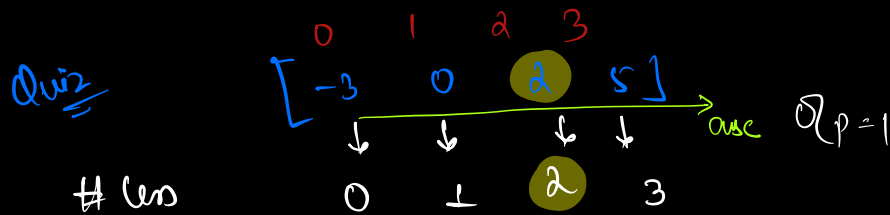
no. of elements lesser than $A[i]$

$== A[i]$



$$O/p = 3.$$

$\therefore$ Can -ve nos. be node integers? ~~XXXX~~



$$O/p = 3$$

Brute force

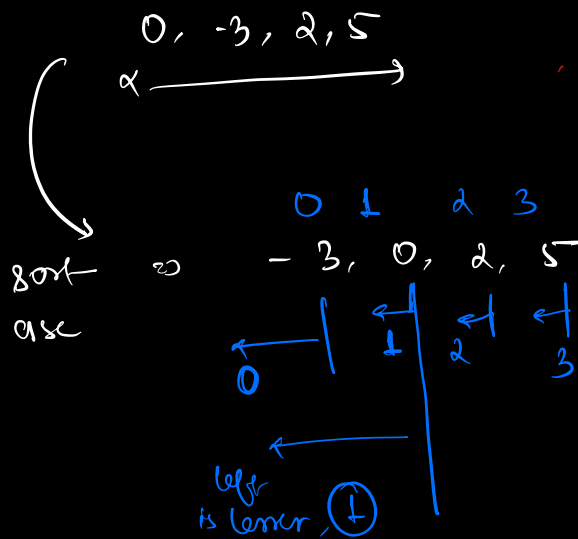
for every element, no. of elements
less than it.

$$TC \Rightarrow O(N^2)$$

$$SC \Rightarrow O(1)$$

pseudo

```
ans = 0
for (i = 0; i < N; i++) {
    count = 0
    for (j = 0; j < N; j++) {
        if (arr[i] > arr[j])
            count++
    }
    if (count == arr[i])
        ans++
}
```



idx is representing
how many elements
are lesser than
current element

```
if (arr[i] == i)
    => noble integer
```

pseudo

```

ans = 0
sort(A) // asc order
for (i=0; i<N; i++) {
    if (i == A[i])
        ans++
}
return ans;

```

TC $\Rightarrow O(N \log N) + O(N)$

$\approx O(N \log N)$

SC $\Rightarrow$ depends on sorting algo

$\Rightarrow$ with duplicates!

Quiz $\Rightarrow$

	0	1	2	3	4	
	-10	1	1	3	100	
	↓	↓	↓	↓	↓	→ ans
# less $\Rightarrow$	0	1	1	3	4	$O/p = 3$

Quiz $\Rightarrow$

	0	1	2	3	4	5	6	7	8	
	-10	1	1	2	4	4	4	8	10	
	↓	↓	↓	↓	↓	↓	↓	↓	↓	→ ans
# less $\Rightarrow$	0	1	1	3	4	4	4	7	8	$O/p = 8$

if, $arr[i] == arr[i-1]$

no. of elements lesser than $arr[i]$

$=$ no. of elements lesser than $arr[i-1]$

if $arr[i] \geq arr[i-1]$

no. of elements lesser than $arr[i]$

i

Step 2) $sort(A)$ // asc order

logic

pseudocode $\rightarrow$ PODO

How sort an array in Asc order by the no. of factors, if factors are same then sort by value

ex $\Rightarrow$ 9 3 10 6 4

factors $\Rightarrow$ 3 2 4 4 3

sort $\Rightarrow$ 3 4 9 6 10 $\Leftarrow$ ans

factors $\Rightarrow$

3 $\downarrow$ 2
4 $\downarrow$ 3
9 $\downarrow$ 3
6 $\downarrow$ 4
10 $\downarrow$ 4

